

Meat and components of meat and the risk of bladder cancer in the NIH-AARP Diet and Health Study

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Background Meat could be involved in bladder carcinogenesis via multiple potentially carcinogenic meat-related compounds related to cooking and processing, including nitrate, nitrite, heterocyclic amines (HCAs), and polycyclic aromatic hydrocarbons. We comprehensively investigated the association between meat and meat components and bladder cancer. Methods During 7 years of follow-up, 854 transitional cell bladder cancer cases were identified among 300,933 men and women who completed a validated food frequency questionnaire in the large prospective NIH-AARP Diet and Health Study. We estimated intake of nitrate and nitrite from processed meat and HCAs and PAHs from cooked meat using quantitative databases of measured values. We calculated total dietary nitrate and nitrite based on literature values. Results The hazard ratios (HR) and 95% confidence intervals (CI) for red meat (HR for fifth compared to first quintile=1.22, 95% CI=0.96–1.54, p-trend=0.07) and the HCA 2-amino-1-methyl-6-phenylimidazo[4,5-b]pyridine (PhIP) (HR=1.19, 95% CI=0.95–1.48, p-trend=0.06) conferred a borderline statistically significant increased risk of bladder cancer. We observed positive associations in the top quintile for total dietary nitrite (HR=1.28, 95% CI=1.02–1.61, p-trend= 0.06) and nitrate plus nitrite intake from processed meat (HR=1.29 95% CI=1.00–1.67, p-trend= 0.11). Conclusions These findings provide modest support for a role for total dietary nitrite and nitrate plus nitrite from processed meat in bladder cancer. Our results also suggest a positive association between red meat and PhIP and bladder carcinogenesis.